

Question	Marking Guidance	Mark	Comments
06.1	0.1;	1	
06.2	Accept answer in the range of 4.7 to 4.9;	1	
06.3	1. (Trexall acts as a) competitive inhibitor OR (Trexall) competes (with folic acid/substrate) for/is able to fit into/binds at <u>active site</u> (on dihydrofolate reductase / enzyme); 2. Less folic acid/substrate attaches OR Fewer enzyme-substrate complexes; 3. Fewer/not enough nucleotides available for DNA replication;	3	1. Reject Trexall and folic acid have the same shape 2. Accept folic acid/substrate is prevented from binding 3. Accept fewer/not enough nucleotides available during interphase/for semi-conservative replication/to add to (all) template strands/for transcription
06.4	Percentage change 1. To allow comparison as tumours may differ in volume/size (at the start of the investigation); Tumour volume 2. (As) tumours may differ in length/width/shape OR (As) volume is (best) indication of the number of cells in tumour;	2	2. Accept 'as tumours are three dimensional' 2. Ignore answers relating to density/thickness

06.5	Answer in the range 32 015.93682 to 32 045 = 2 marks OR $3.20 \times 10^4 = 2$ marks;; Allow 1 mark for correct calculation of volume after treatment in range of 24 011.95261 to 24 034 / 2.40×10^4 Allow 1 mark if student uses diameter throughout instead of radius, in range of 256 127 to 256 361 / 2.56×10^5	2 max	Accept any suitable rounding
06.6	For (the use of 30 mg) 1. There is a significantly greater reduction (in tumour size with 30 mg), as SD (bars) do not overlap; 2. In some cases (with 30 mg) there was a 100% reduction in size/tumours would have been eradicated; Against (the use of 30 mg) 3. There is too much/a lot of variation in effectiveness with 30 mg (in contrast with 20 mg); 4. (No idea of) extra cost of providing 30 mg per week; 5. (Increased risk of) side effects with higher doses;	2 max	Accept converse arguments for all mark points. 1. Accept 'significant difference' for 'significantly greater reduction' 3. Ignore 30 mg has a lot of deviation/large standard deviation' unqualified
06.7	Accept any two suitable suggestions for one mark, eg; Severity/duration of arthritis Current/other medication Type of arthritis Weight/body mass Ethnicity	1 max	Reject age/health as they are directly in the stem Ignore gender/sex Ignore general answers such as diet/activity/lifestyle

06.8	For 1. Pain decreases more with Trexall/Group R compared with the control group/Group S OR Pain decreases by 4.6 with Trexall/Group R and by 2 with the control group/Group S; Against 2. Small sample size/only 12 people/only studied females / effects in males could be different; 3. (Mean score for severity of) pain in control group/Group S is (also) lower; 4. No statistical testing, so do not know if decrease/difference is significant; 5. Pain is (a) subjective (measurement);	3 max	2 max for answer only giving reasons against 1. Ignore numbers stated from Table 3, eg 9.7 to 5.1 and 9.8 to 7.8 3. Could be subsumed within MP1 4. Ignore 'do not know if results are significant' 5. Accept 'patients might lie about pain'
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